

# Amit Nahum

## Data Scientist | Machine Learning & Applied AI

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### PROFILE

I build and evaluate machine learning models, combining hands-on experience in aviation data analysis with predictive modeling, signal processing and offline AI systems. Beginning an M.Sc. in Computer Science at HIT; seeking Data Science and Machine Learning opportunities.

### EXPERIENCE

#### AI Engineer | IDF Reserve Duty | 2025 - Present

- Self-hosted vLLM inference stack with SLM, reranker and embedder models for offline workflows
- LLM evaluation on custom Hebrew and English datasets
- Isolation Forest anomaly-detection workflow with analysis of label and data-quality limitations

#### Junior Data Analyst | IDF | 2020 - 2022

- SQL-based analysis and statistical sampling of aviation safety datasets.

### SELECTED PROJECTS

#### BridgePulse - Signal Processing & ML

Built MFCC feature extraction and Random Forest classification for bridge vibration recordings. Documented mean balanced accuracy of 81.97% across state-grouped allocations; exploratory evaluation on 25 recordings from one bridge, without an untouched final test.

#### License Plate Pricing - Kaggle

Nahum Team placed 2nd out of 694 teams on the final private leaderboard. Public work demonstrates feature engineering, CatBoost regression and SHAP analysis; the public notebook has not been matched to the ranked submission.

#### UFC Fight Prediction

Engineered comparative features from recent fights and career history for CatBoost classification. Chronological evaluation is still required before claiming future-fight performance.

#### JEV Decision Experiments - Applied AI

Compared structured model decisions in flight, driving and chess simulations using matched evaluations, saved traces and replay; results are limited to the documented simulation samples.

### CORE SKILLS

Python, SQL, Statistics, Machine Learning, Feature Engineering, Model Evaluation, Signal Processing, scikit-learn, CatBoost, Docker, vLLM, LLM Inference.

### EDUCATION

#### M.Sc. in Computer Science | HIT | Beginning in 2026; expected completion 2028

Interests: Deep Learning, model optimization and efficient inference.

#### B.Sc. in Computer Science | HIT | 2024 - 2026 | Completed in 2026

Coursework: Statistics & Probability, Database Management (SQL), Operations Research and Information Systems Analysis.